

packages/graph-view/scripts/benchmark-aggregations.mjs

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Overview

A new Node-only script `packages/graph-view/scripts/benchmark-aggregations.mjs` is added to compare the performance of two graph-aggregation strategies used in the dashboard.

Key changes

- **Import** `performance` from `node:perf_hooks` (line 16) to time operations.
- **`makeGraph(layerCount, nodesPerLayer)`** (lines 18-32) builds a synthetic graph with layers and nodes of three complexity levels.
- **`aggregateBefore(graph)`** (lines 35-49) implements the legacy $O(N \times K \times L)$ algorithm that filters nodes per layer with `Array.filter` and `Array.includes`.
- **`aggregateAfter(graph, nodesById)`** (lines 53-73) implements the optimized $O(N + \sum K_i)$ algorithm that looks up nodes via a `Map`.
- **`bench(label, layerCount, nodesPerLayer)`** (lines 75-94) runs both algorithms, logs elapsed time, computes speed-up, and checks that the outputs are identical.
- Four benchmark runs are executed at the bottom of the file (lines 96-99): *small*, *medium*, *large*, and *issue#102 shape*.

Impact

- The script is isolated in the `scripts` folder and does not modify any production code or public APIs.
- It introduces a runtime dependency on `node:perf_hooks` only when the script is executed, so it does not affect the Vite bundle.
- The parity check gives confidence that the refactor preserves functional correctness.

Risks & follow-ups

- **Build exclusion:** confirm that the `scripts` folder is excluded from the Vite build (e.g., via `vite.config.js` or `.gitignore`).
- **Environment guard:** consider adding a check to skip execution in non-development environments to avoid accidental runs in CI.
- **Duplicate ID safety:** the synthetic data guarantees unique node IDs; if reused elsewhere, ensure uniqueness before constructing `nodesById`.
- **Performance sanity:** run the script locally to verify realistic `speedup` values and that parity is always `true`.